

ON THE EXTENSION OF THE PARTIAL ORDER OF GROUPS.*

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1. In his paper "Sur l'extension de l'ordre partiel"¹ E. Szpilrajn has proved that every partial order defined on a set has a linear extension. This general result does not necessarily hold for partially ordered groups, since it may obviously happen that the extended order does not satisfy the group axioms. The principal purpose of the present paper is the demonstration of the same theorem on abelian groups. The theorem will be proved to hold only for groups on which an additional condition is satisfied, a condition which requires that only positive elements have positive natural multiples.

2. We recall that an abelian group G , written additively, is said to be a *partially ordered group*² if a relation $>$ is defined between some pairs of its elements such that the following postulates hold:

- (i) any two of the three relations $a > b$, $a = b$, $a < b$ are contradictory;
- (ii) transitivity: $a > b$ and $b > c$ imply $a > c$;
- (iii) homogeneity: $a > b$ implies $a + c > b + c$ for every c in G .

By the laws (ii) and (iii) the relations $a > b$, $c > d$ may be added to get $a + c > b + d$.

If in addition G satisfies the condition:

- (iv) $na = a + a + \cdots + a \geq 0$ for some positive integer n implies $a \geq 0$,

we say the partial order is *normal*.

If G is such that any two elements a, b are *comparable* in the sense that one of the possibilities $a > b$, $a = b$, $a < b$ does hold, we say G is *linearly ordered*. A linearly ordered group always satisfies condition (iv), for if, under the hypothesis $na \geq 0$, $a \geq 0$ did not hold, we should then have by linear order $a < 0$ implying $na < 0$ for every positive integer n .

We now prove that a group on which a normal partial order is defined has, with the exception of 0, only elements of infinite order. In fact, the normality states that if $na = 0$, or, what is the same, if $na \geq 0$ and $-na \geq 0$, then $a \geq 0$ as well as $-a \geq 0$, that is, $a = 0$.

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¹ *Fundamenta Mathematicae*, vol. 16 (1930), pp. 386-389.

² Our definition is stated in a form which is most convenient for our purpose; cf. C. J. Everett and S. Ulam, "On ordered groups," *Transactions of the American Mathematical Society*, vol. 57 (1945), pp. 208-216.

Suppose that two partial orders P and R are defined on the same group and a relation $a > b$ in P implies $a > b$ in R ; then R will be called an *extension* of P . An extension which defines a linear order on G will be called a *linear extension*.

3. We shall now prove the following lemma.

LEMMA. *If P is a normal partial order on the group G and x and y are any two elements non-comparable in P , then there exists an extension R of P such that $x > y$ in R . Moreover if such an extension may be carried out for any two non-comparable elements, then P is necessarily normal.*³

For the proof assume that P is a normal partial order on G and the elements x and y are not comparable in P . Let us define a relation R as follows.

We put $a > b$ in R if and only if $a \neq b$ and there are two non-negative integers p, q , not both zero, such that

$$(1) \quad p(a - b) \geq q(x - y) \text{ in } P.$$

What we have to show is that by this definition R is a partial order and an extension of P , further $x > y$ in R .

First of all, we note that p is never zero, for otherwise we should have $0 \geq q(x - y)$ in P for a certain positive integer q , whence by (iv) we have $y \geq x$ in P against hypothesis.

α) We begin with verifying condition (i) for R . It is clearly enough to show that $a > b$ in R and $b > a$ in R are contradictory. For assume $p(a - b) \geq q(x - y)$ in P , as well as $r(b - a) \geq s(x - y)$ in P , for some non-negative integers p, q, r, s . By adding r times the first, p times the second inequality, one obtains $pr(a - b) + pr(b - a) \geq (qr + ps)(x - y)$ in P , that is to say, $0 \geq (qr + ps)(x - y)$ in P . If $qr + ps$ does not vanish, by normality we are led to $y \geq x$ in P , a contradiction. On the other hand, if $qr + ps$ is zero, i. e., both q and s vanish, then the inequalities $p(a - b) \geq 0$ in P and $r(b - a) \geq 0$ in P imply by normality $a \geq b$ in P and $b \geq a$ in P , i. e., $a = b$ which is absurd.

β) We proceed now to the proof of the transitivity of R . Assume that $a > b$ in R and $b > c$ in R , that is, for some non-negative integers p, q, r, s , $p(a - b) \geq q(x - y)$ in P and $r(b - c) \geq s(x - y)$ in P . By adding as in α) one gets $pr(a - b) + pr(b - c) = pr(a - c) \geq (qr + ps)(x - y)$ in P . Here pr is not zero and $a = c$ is by α) impossible, so that, by definition, $a > c$ in R , which establishes the transitivity of R .

³ Our proof is in some respect similar to Szpilrajn's proof.

γ) In order to prove the homogeneity for R , take into account that (1) includes only the difference $a - b$ which is equal to $(a + c) - (b + c)$ for every c in G and $a + c = b + c$ is impossible if a and b are different.

δ) R is an extension of P , for if $a > b$ in P , then $a - b > 0$ in P , consequently, for $p = 1$, $q = 0$, condition (1) is satisfied, hence $a > b$ in R .

ε) One sees at once that $x > y$ in R . In fact, for $x = a$, $y = b$ and $p = q = 1$, the relation (1) takes the form $x - y \geq x - y$ in P .

ξ) To complete the proof of the lemma, let us now assume that there is an element g in G such that $ng \geq 0$ in P without $g \geq 0$ in P . Then g and 0 are not comparable in P and we may infer, from the above discussions, that there exists an extension of P in which $g < 0$. This is however absurd, since this would imply $ng < 0$ in R , contrary to the hypothesis $ng \geq 0$ in P .

4. We may prove even the normality for the partial order R defined in 3. Indeed, supposing $na \geq 0$ in R for some positive integer n , i. e., $p(na) = (pn)a \geq q(x - y)$ in P , we are led at once to the result $a \geq 0$ in R .

What has been proved shows that the extended partial order R may again be extended to another partial order S , in which two prescribed elements non-comparable in R become comparable, etc.

If, in general, $P_1, P_2, \dots, P_\tau, \dots$ is a well-ordered chain of partial orders such that each of them is some extension of the preceding ones, then the union of the chain may be defined to be a partial order P such that $a > b$ in P if and only if $a > b$ in P_τ holds for one and hence for all subsequent subscripts τ . There is no difficulty in establishing that P is normal if all P_τ are normal.

Hence, as a simple consequence of Zorn's lemma⁴ we get that in the set of all normal partial orders defined on the group G which are extensions of P , there are maximal orders M , that is, orders which have no proper extension. By our Lemma this can happen only in case any two elements are comparable in M , that is to say, M is a linear order. Thus we have, immediately, the result:

THEOREM 1. *For every normal partial order P defined on G and every two elements x, y non-comparable in P , there exists a linear extension L_{xy} with the property that $x > y$ in L_{xy} .*

⁴ M. Zorn, "A remark on method in transfinite algebra," *Bulletin of the American Mathematical Society*, vol. 41 (1935), pp. 667-670.

5. Let $\mathfrak{S} = \{P_1, P_2, \dots, P_\tau, \dots\}$ be any set of partial orders, each defined on the same group G . We define a new partial order P on G as follows. For any two elements a, b we put $a > b$ in P if and only if $a > b$ in every partial order P_τ in the set $\mathfrak{S}$. It is readily seen that P is again a partial order, moreover, P is normal if all partial orders of $\mathfrak{S}$ are normal. The partial order P is said to be the *product* of the P_τ or to be *realized* by the set $\mathfrak{S}$ of partial orders, written $P = \Pi P_\tau$.

THEOREM 2. *A partial order P defined on a group G may be realized by a certain set of linear orders if and only if P is normal.*

The necessity is obvious, since a linear order, and hence every product of linear orders, is normal. On the other hand, if P is not itself linear, then take to any pair of elements x, y non-comparable in P the corresponding linear extensions L_{xy} and L_{yx} described in Theorem 1. It is easily seen that these linear orders realize P .

If by the dimension⁵ of a partial order P we mean the least cardinal number $\mathfrak{w}$ such that P may be realized by $\mathfrak{w}$ linear orders, then we can reformulate Theorem 2:

THEOREM 3. *A partial order P defined on a group has a dimension if and only if P is normal.*

This theorem states, for example, that each commutative group which is lattice-ordered in the sense of G. Birkhoff,⁶ has a well-defined dimension.

6. When we start with an abelian group on which the relation $>$ is defined for no pair of elements, then, by applying Theorem 2 we come to a theorem due to F. Levi.⁷

THEOREM 4. *In a commutative group a linear order may be defined, if all of its elements, except 0, are of infinite order.*

Indeed, a group in which no partial order is defined is normal if and only if it has no element of finite order other than 0.

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⁵ This definition is due to Ben Dushnik and E. W. Miller, "Partially ordered sets," *American Journal of Mathematics*, vol. 63 (1941), pp. 600-610.

⁶ G. Birkhoff, "Lattice-ordered groups," *Annals of Mathematics* (2), vol. 43 (1942), pp. 298-331. Lemma 3 of §9 states that a lattice-ordered abelian group is always normal.

⁷ F. Levi, "Arithmetische Gesetze im Gebiete diskreter Gruppen," *Rendiconti Palermo*, vol. 35 (1913), pp. 225-236.